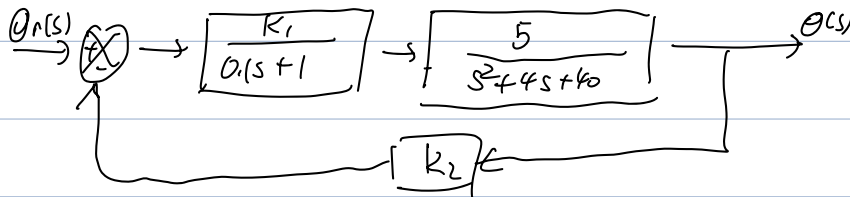


2021

Q2.

(a) If no wind disturbances:



$$\frac{\theta(s)}{\theta_r(s)} = \frac{\frac{K_1 \cdot 5}{(0.1s+1)(s^2+4s+40)}}{1 + \frac{K_1}{0.1s+1} \cdot \frac{5}{s^2+4s+40} \cdot K_2} = \frac{50K_1}{(s+10)(s^2+4s+40) + 50K_1K_2}$$

$$= \frac{50K_1}{s^3 + 14s^2 + 80s + 50(K_1K_2 + 8)}$$

(b). When $\theta_r(s) = \frac{1}{s}$

$$\theta_{ss} = \lim_{s \rightarrow 0} s \cdot \theta(s) \cdot \frac{1}{s} = \frac{50K_1}{50(K_1K_2 + 8)}$$

To let $\theta_{ss} = 1$, we have:

$$\frac{50K_1}{50(K_1K_2 + 8)} = 1$$

$$\Rightarrow K_1 = K_1K_2 + 8$$

$$\theta(s) = \frac{4}{s^2} \cdot \frac{(s+10)(s^2+4s+40)}{s^3 + 14s^2 + 80s + 50K_1}$$

$$\dot{\theta}_{ss} = 4 \cdot \frac{400}{50K_1} = 0.1$$

$$K_1 = 320$$

(c). When $\theta_r(s) = \frac{4}{s^2}$

$$\theta(s) = \frac{4}{s^2} \cdot \frac{50K_1}{s^3 + 14s^2 + 80s + 50K_1}$$

$$W(s) = s \theta(s) = \frac{4}{s} \cdot \frac{50K_1}{s^3 + 14s^2 + 80s + 50(K_1K_2 + 8)} - W(0)$$

$$\text{It's supposed } W_{ss} = \lim_{s \rightarrow 0} s \cdot \theta(s) - W(0) = 4 - 0.1 = 3.9 \text{ rad/s}$$

(1.10)

(1.10)

Therefore: $\omega_{ss} = \lim_{s \rightarrow 0} s \theta(s) = \frac{4 \times 50 K_1}{50(K_1 K_2 + 8)} = \frac{4 K_1}{K_1 K_2 + 8} = 3.9$

$$\frac{K_1}{K_1 K_2 + 8} = \frac{3.9}{4}$$

$4 - \omega(0) = 3.9$
 $\Rightarrow \omega(0) = 0.1$
 $K_1 = 64$ $K_2 = 7/8$

$K_1 K_2 + 8 = 64$

$$\begin{aligned} \theta_w(s) &= \frac{4}{s} \cdot \frac{s^3 + 14s^2 + 80s}{s^3 + 14s^2 + 80s + 50K_1} \\ &= \frac{4(s^2 + 14s + 80)}{s^3 + 14s^2 + 80s + 50K_1} \end{aligned}$$

$\lim_{s \rightarrow 0} s \theta_w(s) = 0$

$$\theta(s) = \frac{4}{s^2} \cdot \frac{3200}{s^3 + 14s^2 + 80s + 3200}$$

4×3200

$s^2(s + 18.82)(s - 2.40 + 12.82j)$

$(s - 2.40 + 12.82j)$

$$= \frac{A}{s^2} + \frac{B}{(s + 18.82)} + \frac{C}{(s - 2.40)^2 + 12.82^2}$$

$$\theta(t) = At + Be^{-18.82t} + C \cdot e^{2.40t} \sin 12.82t$$

$$\omega(t) = A + Be^{-18.82t} + (\dots)$$

$\omega(0) \neq 0$

$C e^{2.40t} \cos 12.82t$